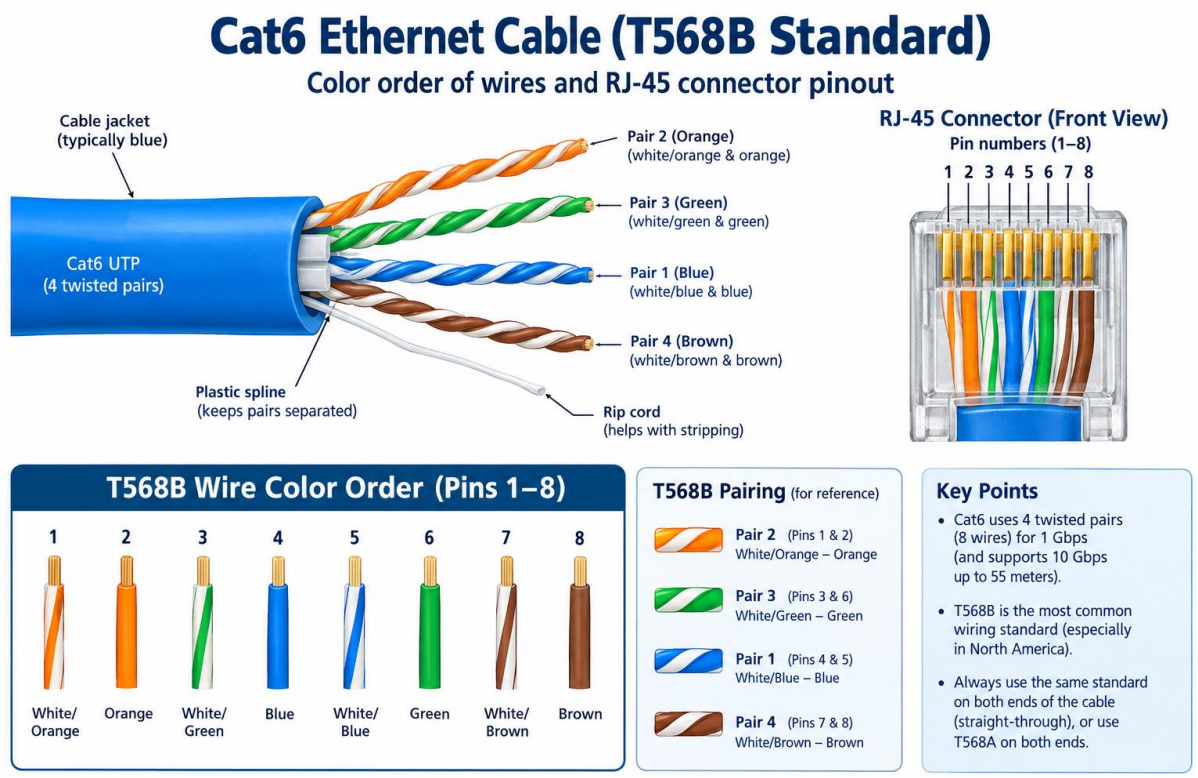
Network cable and connector

Task 1:- Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



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Task 2:- List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Network cable type	Real-world application/device	Why it is used
1. Cat5e Ethernet	Zomato POS terminal in a restaurant	Provides a reliable wired connection to the restaurant's network for processing orders and communicating with billing systems. It supports speeds up to 1 Gbps.
2. Cat6 Ethernet	Wi-Fi router connected to a switch or modem	Supports speeds up to 1 Gbps at 100 meters and can support 10 Gbps over shorter distances, making it suitable for fast local network connections.
3. Fiber optic cable	IPL live-streaming server in a data center	Carries large amounts of data at high speeds over long distances with low signal loss, helping deliver live cricket video streams to viewers.

Task 3:- Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Procedure:

1. Take an Ethernet cable and remove about 2–3 cm of its outer jacket using a wire stripper.
2. Separate the four twisted pairs and straighten the eight individual wires carefully.
3. Arrange the wires according to the T568B standard in the following order:
    1. White/Orange
    2. Orange
    3. White/Green
    4. Blue
    5. White/Blue
    6. Green
    7. White/Brown
    8. Brown
4. Hold the wires in the correct order and trim their ends evenly.
5. Insert all eight wires into the RJ-45 connector, ensuring that they reach the front of the connector.
6. Make sure the outer cable jacket extends into the connector so that the strain relief can hold it securely.
7. Insert the connector into the appropriate slot of the crimping tool and squeeze the handles firmly.
8. Repeat the process at the other end of the cable using the same T568B standard for a straight-through cable.
9. Connect both ends to a LAN cable tester and verify that all eight pins are connected correctly.

Two Common Mistakes to Avoid:

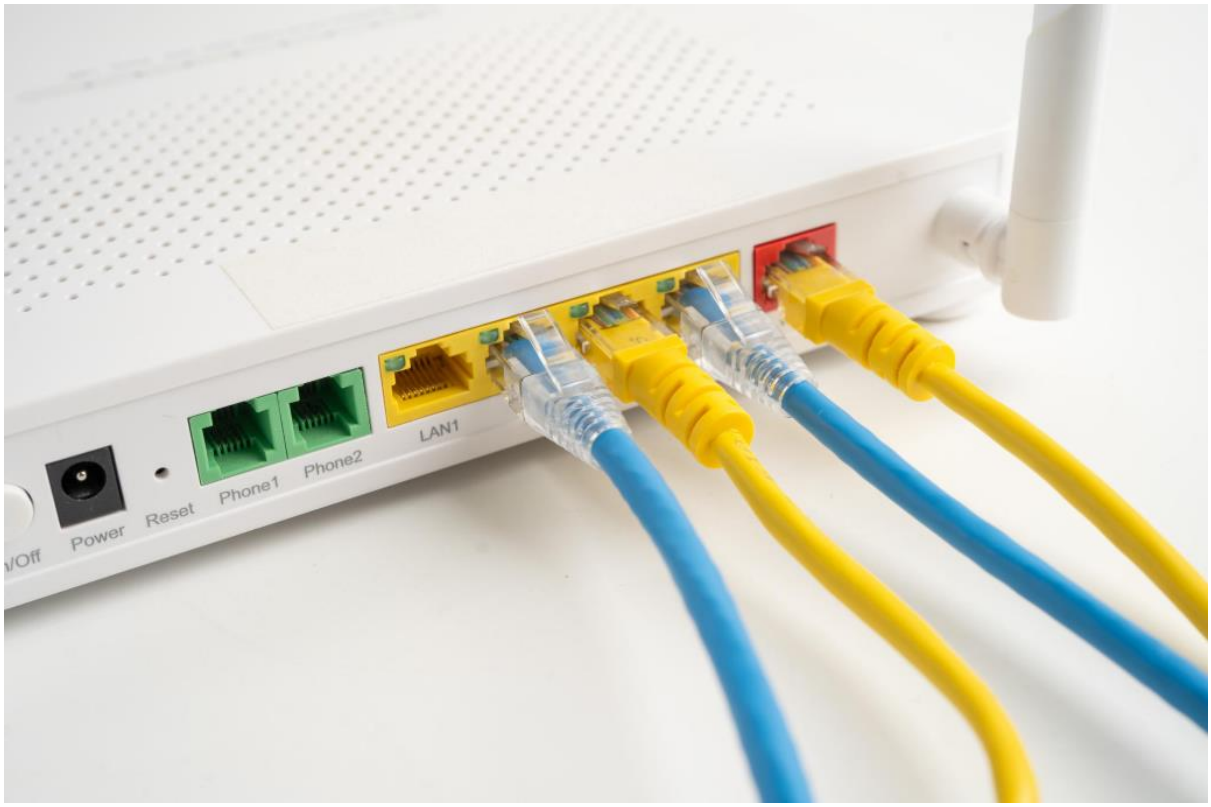
1. **Incorrect wire sequence:** Arranging the wires in the wrong color order can cause wiring faults or connection failure. Always follow the T568B standard.
2. **Improper wire insertion:** If the wires are not fully inserted or the cable jacket is not secured inside the connector, the connection may become unreliable.

Result: The RJ-45 connector was crimped onto the Ethernet cable using the T568B standard. The cable can be used for network communication after successful testing.

Task 4:- Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Cable type	Connector type	Application	Reason for selection
Cat6 Ethernet cable	RJ-45	Connecting gaming consoles and network switches to the router	Provides reliable, low-latency network connectivity and supports speeds up to 1 Gbps at 100 meters.
HDMI 2.1 cable	HDMI connector	Connecting gaming consoles and media players to TVs or monitors	Supports high-resolution video and high refresh rates, suitable for smooth gaming and IPL screening.
Fiber optic cable	LC or SC connector	Connecting the main network switch to a distant network switch or server	Supports high bandwidth over long distances with minimal signal loss and immunity to electromagnetic interference.
Cat6a Ethernet cable	RJ-45	Connecting high-performance gaming PCs or servers to the network	Supports 10 Gbps Ethernet up to 100 meters, providing additional capacity for demanding network traffic.

Task 5:- Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



Type of cable: Ethernet cable (Cat5e or Cat6).

Connector type: RJ-45 connector.

Description: The photograph shows blue Ethernet cables connected to the LAN ports of a Wi-Fi router. The cables have transparent plastic RJ-45 connectors at their ends.

Devices connected: The Ethernet cables connect the router to network devices such as computers, switches, or other networking equipment. The router provides access to the local network and internet.

Why it is used: Ethernet cables provide a stable, high-speed wired connection with low latency. They are useful for online gaming, video streaming, and transferring data reliably.